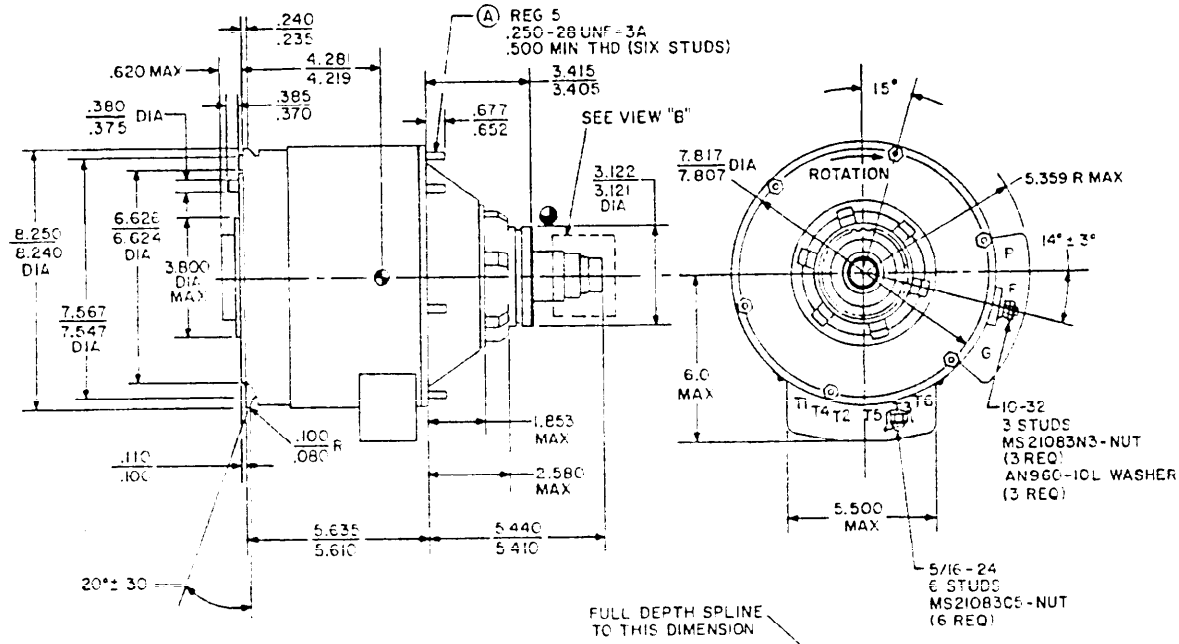
MILITARY SPECIFICATION SHEET
GENERATOR SYSTEM, 30 KVA, 400 HERTZ ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS TYPE

THIS SPECIFICATION IS APPROVED FOR USE BY NAVAL AIR SYSTEMS COMMAND, DEPARTMENT OF THE NAVY, AND IS AVAILABLE FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

THE COMPLETE REQUIREMENTS FOR PROCURING THE GENERATOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF MIL-G-21480.



DIA MARKED ⊕ TO BE CONCENTRIC WITH DIA MARKED ⊗ WITHIN .001 T.I.R.

DIA MARKED ⊗ TO BE CONCENTRIC WITH DIA MARKED ◕ WITHIN .002 T.I.R.

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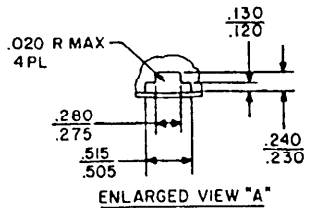


FIGURE 1. M21480/21-2 GENERATOR

FSC 6115

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GENERATOR DATA	
RATED VOLTAGE	120/208
RATED OUTPUT	30 KVA
PHASE	3
FREQUENCY	380 - 420 HZ
SPEED RANGE	7600 - 8400 RPM
MAX SPEED FOR REGULATION	8800 RPM
OVERSPEED	10,600 RPM
RATED POWER FACTOR	0.75 LAG TO 1.0
(A) EFFICIENCY, MIN AT RATED LOAD	84.5 % at unity P.F.
MAX WEIGHT	52 LBS

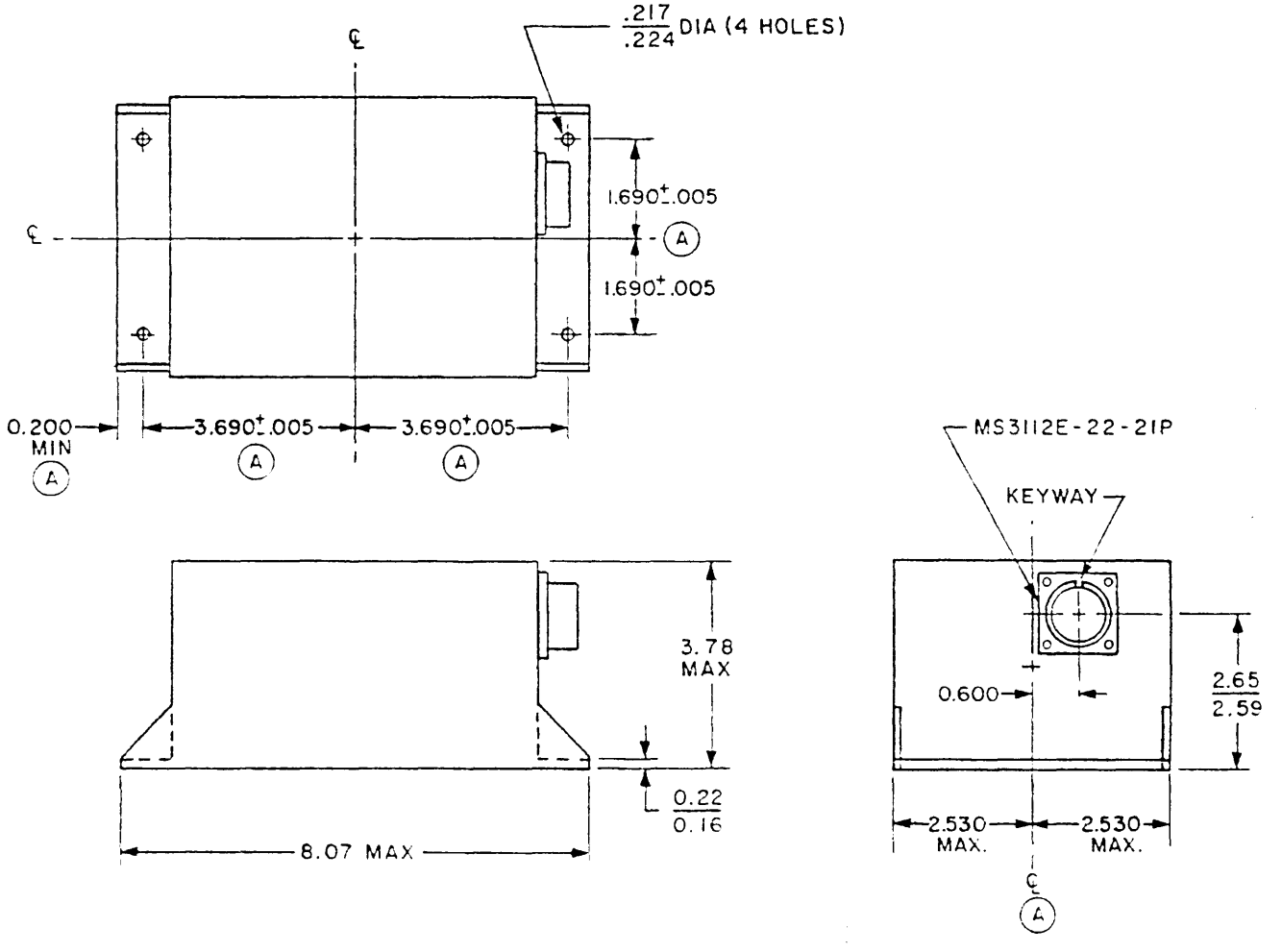
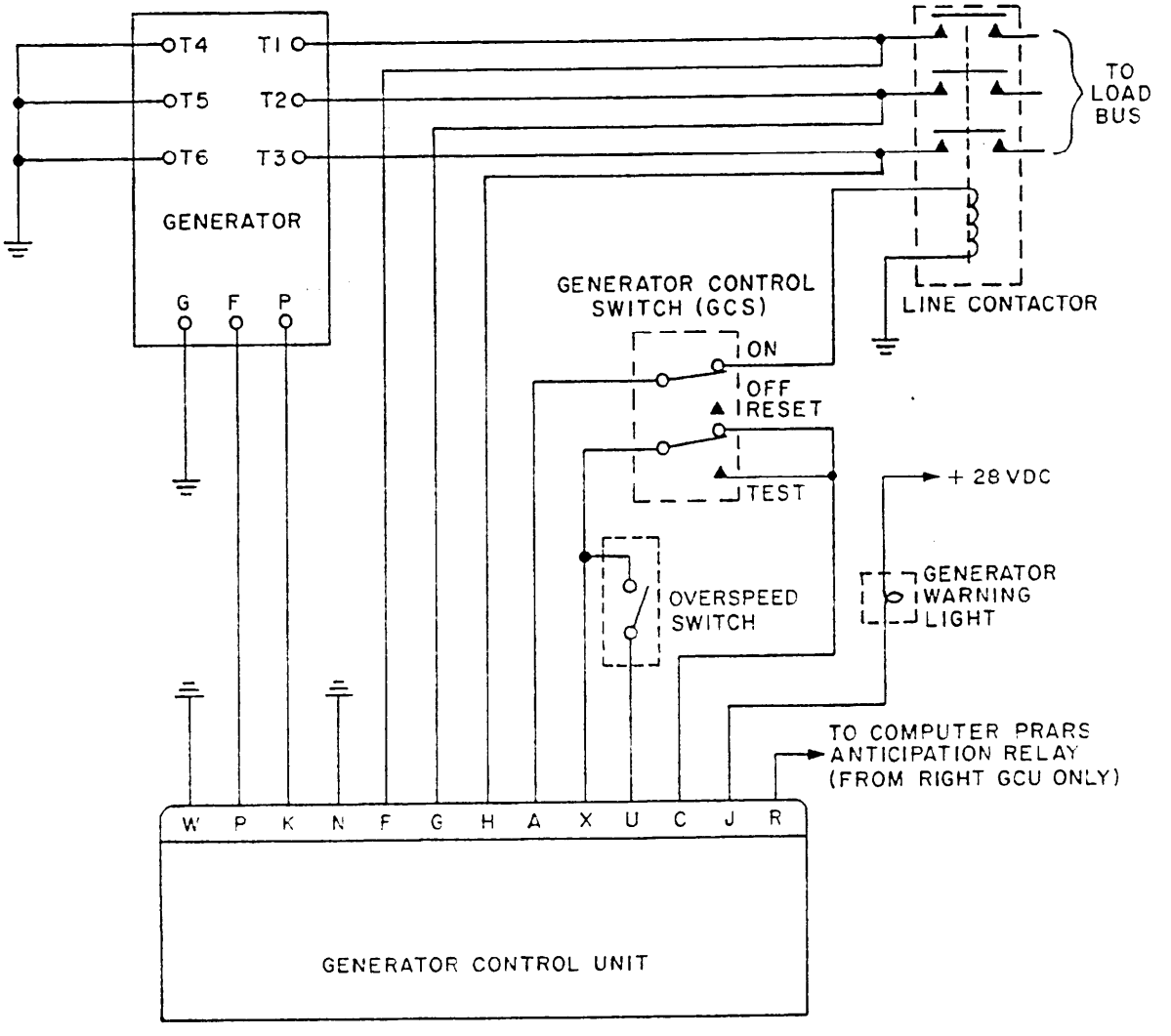


FIGURE 2. M21480/21-3 Generator control unit, max. wt. 4.5 lbs.

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NOTE: Components enclosed in dotted lines are shown for interconnections only and are not furnished under this drawing. Connecting leads shown shall be provided by the aircraft manufacturer.

FIGURE 3. Test circuit for M21480/21-1 system.

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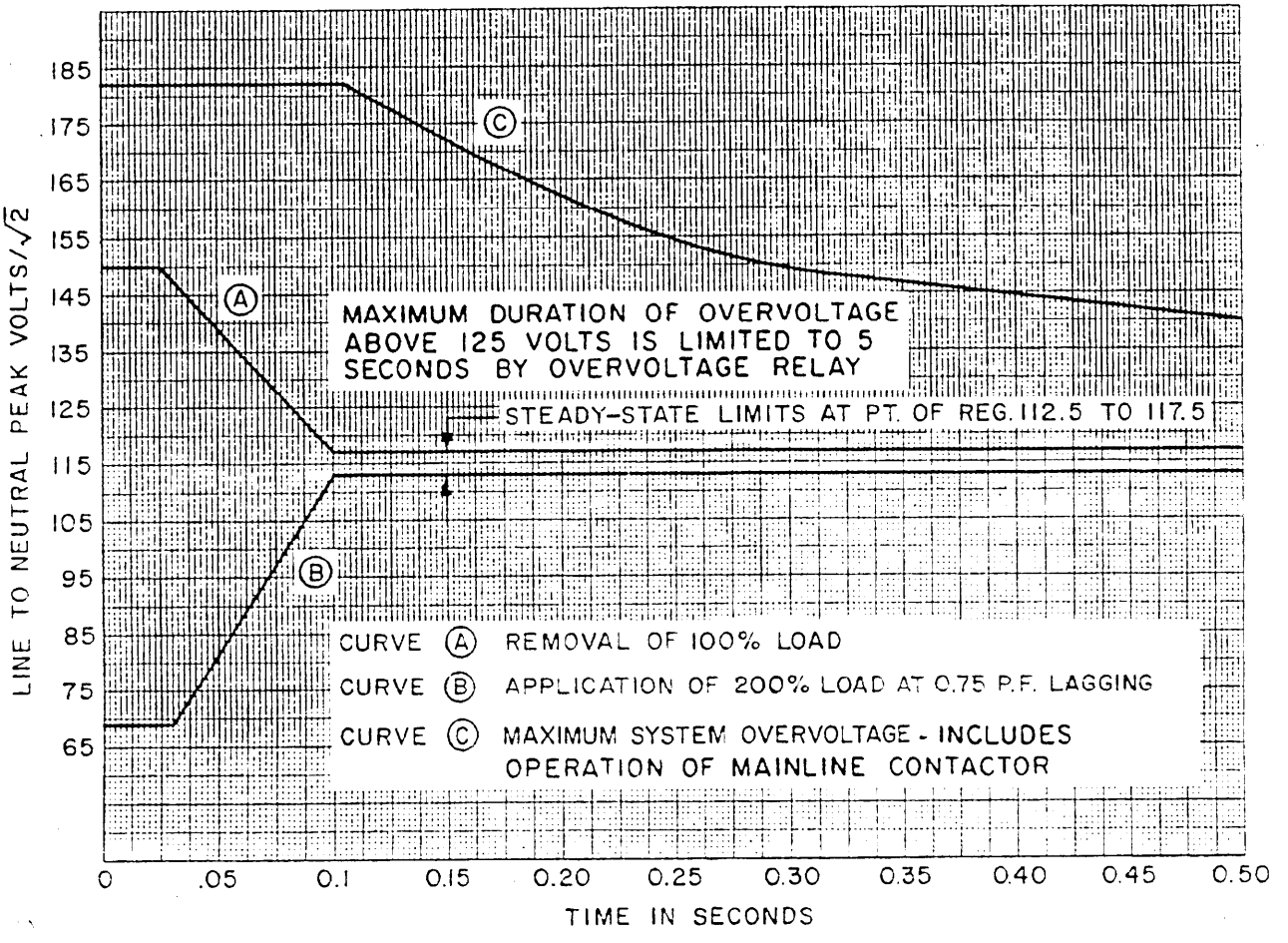


FIGURE 4. Transient, steady-state and overvoltage limits.

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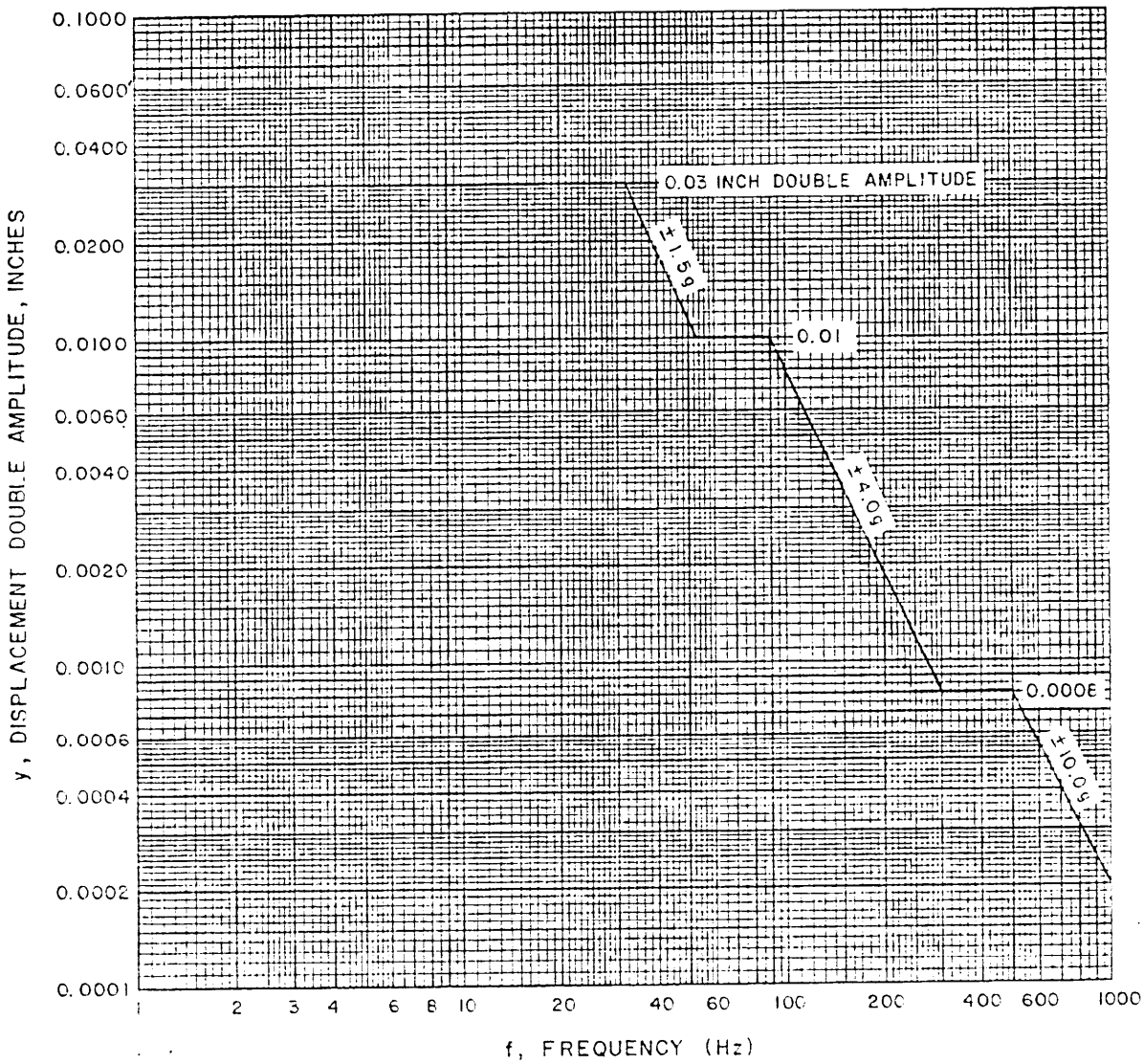
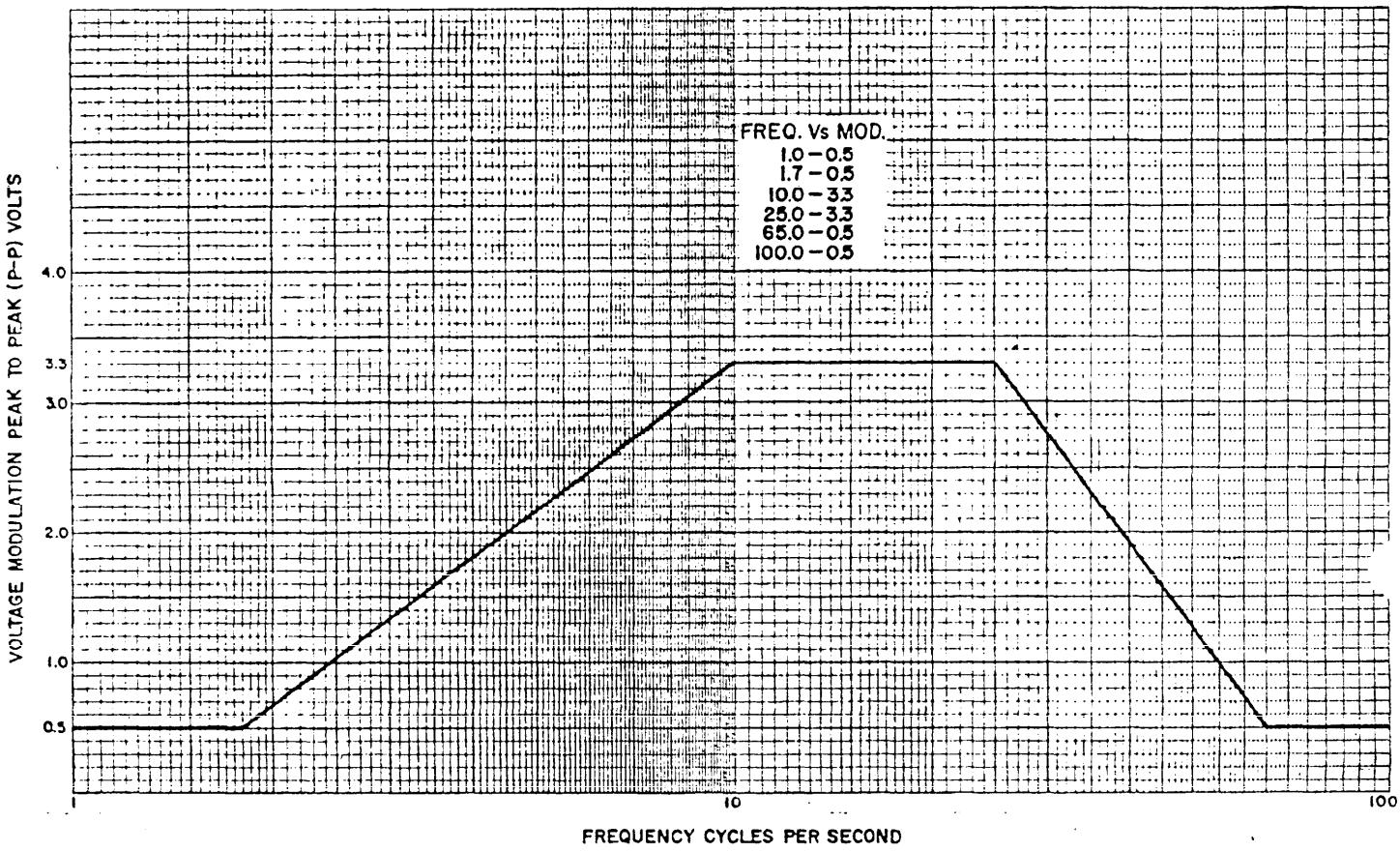


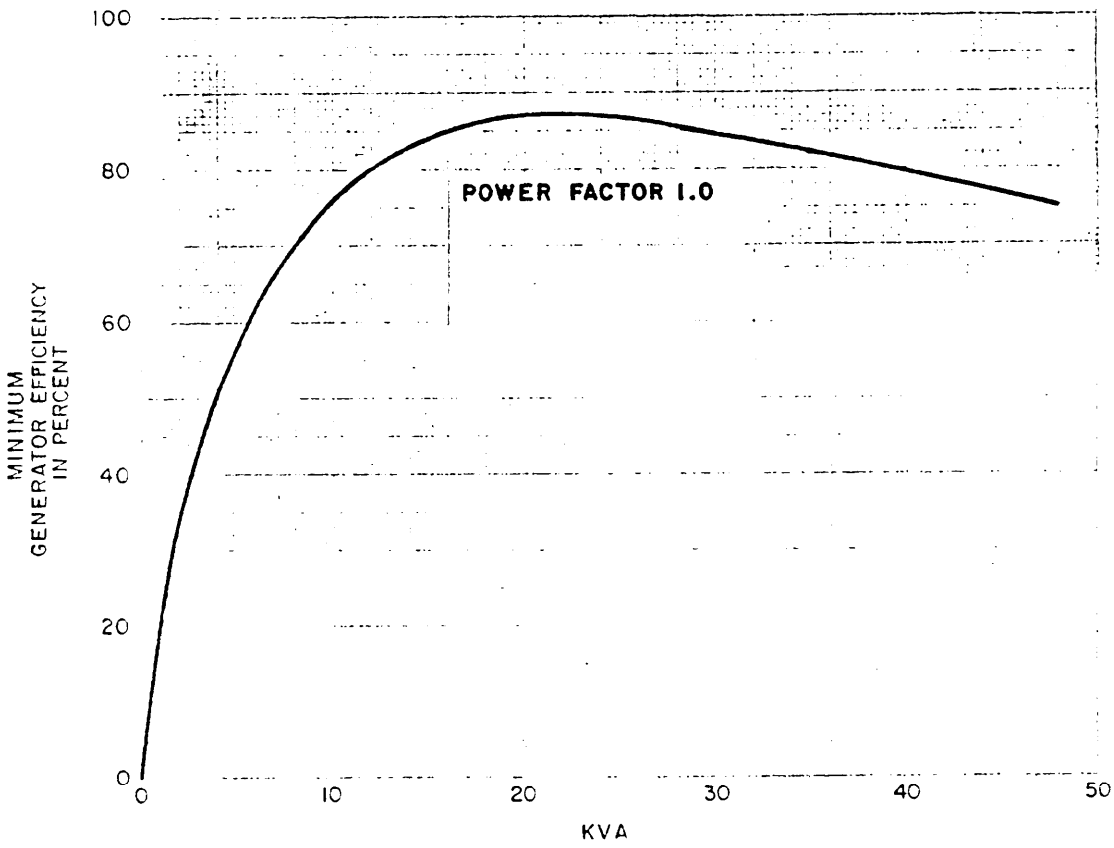
FIGURE 5. Vibration test curve for generator.

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(A) FIGURE 6. Frequency characteristics for voltage modulation limits.

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(A) FIGURE 7. Generator efficiency.

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REQUIREMENTS

The requirements of MIL-G-21480 apply with the following exceptions to existing paragraphs:

Delete the following paragraphs: 3.4.9.2, 3.4.9.6.2, 3.4.10.2, 3.4.11, 3.4.14, 3.4.15, 3.4.16 through 3.4.16.5, 3.5.7.6, 3.5.9, 4.4.1.4, 4.4.2.3.2, 4.5.22.1 through 4.5.22.3.4, 4.5.24.1 through 4.5.24.3.4, 4.5.25.1 through 4.5.25.2.1, 4.5.26, 4.5.27, 4.5.28, 4.5.32, 4.5.33 through 4.5.33.2, 6.2.1.1.

3.3.3.3 Magnesium alloy parts. Delete and add: Magnesium parts shall not be used.

Add: 3.3.5 Explosion and fire hazard. When operated under any of the conditions specified herein, the equipment shall not cause ignition of an explosive mixture.

3.4.2.1 Temperature and altitude. Delete and add: The generator shall meet the temperature-altitude requirements of MIL-E-81910 for blast cooled equipment. When operated under self-cooled conditions, the generator shall be capable of providing 100 percent load at any altitude up to 6000 feet. No performance is required over 50,000 feet.

3.4.2.2 Humidity. Add: "No operation required during exposure time."

3.4.2.3 Salt spray. Delete and add: As required by paragraphs 4.5.20.1 and 4.5.20.2 of this specification sheet.

3.4.2.4 Fungus. Delete and add: As required by paragraph 4.5.21 of this specification sheet.

Add: 3.4.2.8.1 Flight-maneuver forces. The components and their mounting provisions shall withstand the following conditions without permanent deformation or failure:

Add: 3.4.2.8.1.1 Identification of forces. The direction of the forces quoted herein shall be identified by the following axes.

Unit Orientation (Direction)	Corresponding Unit Axis
Longitudinal (fore and aft)	X
Transverse (sideward)	Y
Downward or upward	Z

The unit is in the normal horizontal position when the Z-axis coincides with the direction of gravity.

Add: 3.4.2.8.1.2 Linear acceleration (acting at the center of gravity of the unit).

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a. 11 g's acting vertically upward and 7 g's acting vertically downward (along the Z-axis).

b. 5 g's acting forward and 5 g's aft (along the X-axis).

c. 2 g's acting sideward in either direction (along the Y-axis).

Add: 3.4.2.8.1.3 The components or their mounting provisions shall not fail when subjected to static loads equivalent to 1.5 times the loads specified in 3.4.2.8.1.2.

The generator will be mounted with the center line of its shaft parallel to the aircraft engine horizontal center line and oriented in the forward-aft direction. Orientation of mounting of the other components shall be based on the aircraft installation.

Add: 3.4.2.8.2 Gyroscopic moments. At maximum in-flight speed, the components and their mountings shall withstand a gyroscopic moment imposed by steady angular velocities, as tabulated below, for a period of 30 seconds.

Axis	Angular Limit (Degrees)	Velocities (Degrees/Second)	Acceleration (Rad/Sec^2) Landing	In-Flight
Pitch	None (360)	50	15	12
Roll	None (360)	160	15	30
Yaw	±15	20	15	0.7

Add: 3.4.2.8.3 Demonstration of loads. Demonstration of any of the loads specified in 3.4.2.8.1 or 3.4.2.8.2 is not required.

3.4.7 Manual system control. Add: The test function is required.

3.4.8 System capacity (rating). Delete and add: The nominal rating of the generator system shall be 30 KVA at 8000 rpm at all power factors from 1.00 to 0.75 lagging. The nominal output of the generator shall be 120/208 volts, 400 Hz, in a three-phase, four-wire grounded neutral system. The rated line current shall be 83.3 amperes. The generator shall be sized for a 5 volt drop in each feeder at rated load between the generator terminals and the point of regulation. The voltage limits of figure 4 of this specification sheet pertain.

3.4.8.2 Overload capacity. In third line change "5 minutes" to "2 minutes." Add: With the specified blast cooling the generator system shall be capable of supplying 38 KVA for one hour at any rated power factor. The 150 percent rated load is not required with self-induced cooling.

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Add: 3.4.8.4 Overspeed. The generator rotor shall be built to withstand being rotated up to a maximum speed of 10,600 rpm. Operation for 30 seconds at 9000 rpm for 1200 times (the anticipated number of engine starts during 1000 hours of time between overhauls) shall not adversely affect the life or the operation of the generator. The generator is not normally energized under the above conditions. The stator shall contain the rotor parts in the event that the overspeed circuit malfunctions. The overspeed protection incorporated in the drive will normally prevent generator speeds in excess of 9200 ± 400 rpm. Maximum speed for regulation shall be 8800 rpm.

3.4.9.3 Electrical connections. Add: The generator system shall be so designed that upon removal of any connector plug the generator and generator control unit shall be de-energized.

3.4.9.4 Direction of rotation. Change: "counterclockwise" to "clockwise."

3.4.9.6 Cooling. Add: The generator shall supply 100 percent load continuously at any altitude up to 6000 feet in a self-induced cooling mode without change in the rated life of the generator.

3.4.9.6.1 Air cooling. Add: The generator shall supply rated load at any rated power factor with the maximum cooling airflow limits of figure 5 of MIL-G-21480. Maximum pressure drop at sea level shall not exceed 11 inches of water.

3.4.9.6.1.1 Altitude rating chart. In second line change "(10,000 hour life)" to "(1000 hour life)."

Delete paragraph 3.4.9.7 and replace with following.

3.4.9.7 Normal life. The normal useful life of the equipment with the specified allowable overhaul activity shall consist of the following:

a. An elapsed life of 10 years.

b. An operation life of 10,000 hours.

c. A normal operating period of 4 hours between stops.

d. A shelf life of at least 5 years.

Add: 3.4.9.9 Leakage. The generator shall perform satisfactorily and suffer no adverse effects from lubricating oil per specification MIL-L-23699 or MIL-L-7808, that may reach the generator through its seal. The seal shall not permit oil leakage in excess of 1.0 cc per hour at ambient pressure and temperature.

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Add: 3.4.9.10 Lubrication. The generator anti-drive end bearing shall not require the addition of lubricant during 1000 hours of operation. The generator drive-end bearings will be supplied lubrication from the constant speed drive gearbox.

3.4.10 Control component design. Delete and add: The GCU shall be composed of replaceable function modules where practical, and shall protect the power system against overvoltage, undervoltage, underfrequency, overfrequency and recycling under fault conditions.

3.4.10.3 Life. Delete and add: System control components shall have a minimum of 5000 hours operational service life.

3.4.10.5 Modular construction. Delete and add: The components can utilize potting compounds in mounting of the internal components, such that individual components are not accessible for adjustment or replacement. The basic assembly may be subdivided into individual modules or packages which may be completely encapsulated in the potting compound.

3.4.10.6 Potting compounds. Add: Proprietary compounds may be used providing they meet the environmental requirements of the applicable equipment specifications.

3.4.12.1 Terminals. Add: The terminal markings shall be incapable of being reversed.

3.4.13.1 Nomenclature. Delete and add: An appropriate nameplate in accordance with MIL-STD-130 shall be attached securely to each major component of the generating system in a manner that minimizes corrosion in the presence of salt atmosphere. The nameplates shall be so located on each component that they may be readily seen, and that failure of the component will not be caused when the plate is detached. The maximum size of each component nameplate shall be 2.5 by 3.5 inches, and the nameplate shall incorporate the following data in addition to that required by MIL-STD-130 for "units, sets, and all other items of military property":

a. Government acceptance stamp

b. Date of Government acceptance

c. Serial number

NOTE: Items a. and b. may be stamped on unit near nameplate.

3.4.17 Reliability. The system MTBF (mean time between failure) shall not be less than 1000 hours.

3.5.1 Voltage limits. Change "figure 3" to "figure 4 of this specification sheet."

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3.5.2.1 Amplitude. Change "1.0 volts" to "3.3 volts."

3.5.2.2 Frequency characteristics of the voltage modulation. Change "figure 4" to "figure 6 of this specification sheet."

3.5.4 Waveform. Delete and add: The crest factor for each phase voltage shall be 1.41 ± 10 percent. No single harmonic shall exceed 3 percent of the fundamental.

3.5.5 Integral control power. Replace last sentence with "Power for external use, including the line transfer contactor shall have the following characteristics:

a. Minimum voltage: 20 volts DC

b. Maximum voltage, continuous: 30 volts DC

c. Maximum current: Continuous .8 amperes, Transient 1.2 amperes"

3.5.7.1 False trips. Delete and add: No tripping of the generating system shall be caused by buildup, load switching, input speed variations within the limits defined in 3.5.7.1.1, and external fault applications or removals.

Add: 3.5.7.1.1 Input speed variations. The generator shall operate as required by this specification sheet with the following input speed variations.

a. Steady-state, within ±1.0 percent.

b. Normal transient, within ±5.0 percent, with a 2-second recovery to ±1.0 percent.

c. Any combination of transients, within ±10.0 percent, with a 2-second recovery to ±1.0 percent (and shall not cause protective devices to function).

Add: 3.5.7.1.2 Fault capacity. The generator system components shall withstand the following fault conditions during operation without sustaining damage:

a. A minimum of 300 percent single-phase line-to-neutral, line-to-line, and three-phase fault current at the output of the line transfer contractor for a minimum of five seconds.

b. Any fault, as defined in 6.2.1, at any point in the aircraft system.

Feeder impedance limits shall be specified by the generator manufacturer.

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3.5.7.2 Overvoltage. Delete and add: The system shall provide for the generator to be disconnected from the load bus and de-energized before the voltage exceeds the limits defined by the maximum system overvoltage curve of figure 4 of this specification sheet.

3.5.7.3 Undervoltage. Delete and add: Undervoltage protection shall be provided by a static element which functions to disconnect the system from the load bus and de-energize the generator within a period of from 1.8 to 2.4 seconds after any phase voltage (line-to-neutral) falls and remains below 90 volts. This static element shall not function above 103 volts. Manual reset of this protective element shall be possible by putting the generator switch (see paragraph 3.4.7) in the "OFF RESET" position (then to "ON"). Reset shall be automatic when the system is shut down. After either automatic or manual reset, the undervoltage protection element shall provide for the system to be automatically reconnected to the load bus within 2 seconds after the lowest phase voltage exceeds and remains above 105 volts.

3.5.7.4 Underfrequency. Delete and add: Underfrequency protection shall be provided by a static element which functions to disconnect the system from the load bus and de-energize the generator within a period of from 1.8 to 2.4 seconds after frequency falls below and remains below 370 Hz. This static element shall not function above 380 Hz. Manual reset shall be possible by putting the generator switch (see paragraph 3.4.7) in the "OFF RESET" position (then to "ON"). Reset shall be automatic when the system is shut down. After either automatic or manual reset the underfrequency protection shall provide for the system to be automatically reconnected to the bus within 2 seconds after the frequency exceeds 386 Hz.

3.5.7.5 Overfrequency. Delete and add: Overfrequency protection shall be provided by a static element which functions to disconnect the system from the load bus within a period of from 1.8 to 2.4 seconds after the frequency rises above and remains above 440 Hz. This static element shall not function below 430 Hz. Manual reset of this protective element shall be possible by putting the generator switch (see paragraph 3.4.7) in the "OFF RESET" position (then to "ON"). Reset shall be automatic when the system is shut down. The system shall be reconnected to the load bus within 2 seconds after the frequency falls and remains below 420 Hz.

3.5.8 Electromagnetic interference. Delete and add: The system shall be subjected to an electromagnetic interference test in accordance with MIL-E-81910 except that the ten microfarad feedthrough capacitor shall be omitted.

Add: 3.5.10 Efficiency. The efficiency of the generator, as determined at rated speed, from 50 percent to 150 percent rated load, and 1.00 power factor, shall meet the requirements of figure 7. Losses due to the voltage regulator shall be considered in the generator efficiency.

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Control panel losses shall not exceed 75 watts total, under normal operating conditions. Test in accordance with paragraph 4.5.9 of this specification sheet.

Add: 3.5.11 Fault torque. With a 30 KVA, 0.75 lagging to 1.0 power factor load connected to the generator, there shall be no value of line-to-line or line-to-neutral fault impedance that will cause the constant speed drive to deliver more than 52.2 foot pounds of torque.

3.7 Screw threads. Delete and add: Screw threads shall conform to the requirements of specification MIL-S-7742.

Add: 3.7.1 Special screw threads and retainers. Special screw thread forms for locking devices shall be acceptable where load requirements warrant their use. Retaining rings shall not be used.

Add: 3.7.2 Inserts. Threads in aluminum alloy for use with fittings having a thread diameter of less than .75 inch and subject to removal for routine maintenance purposes shall be provided with inserts.

Add: 3.7.3 Coating threaded parts. When aluminum or aluminum alloy threaded parts are used, they shall be treated at the time of assembly with anti-seize compound conforming to specification MIL-T-5544 or shall be surface-treated in accordance with specification MIL-C-5541.

3.8 Safety wiring and staking. Delete and add: All threaded parts shall be securely locked by safety wiring, self-locking nuts, cotter pins, or other approved methods. Safety wire shall conform to MS20995. Self-locking nuts shall be in accordance with MS33588. Staking shall be avoided wherever possible.

Add: 3.9 Friction-retained pins. Friction-retained pins without auxiliary positive retention means shall not be used. Auxiliary positive retention means that the pin will be mechanically held in place in the event there is no friction between the pin and the hole. Trapping the pin so it cannot fall out is acceptable, provided that the enclosure used to trap the pin has no motion with respect to the pin. Peening, staking, or safety wiring of the pin is not acceptable.

4.2.1.1 Qualification. Delete and add: Qualification tests shall consist of all those tests listed on table I of this specification sheet. Except for the endurance test, 4.5.14, documentation of a test by similarity to other qualified components that have successfully passed such a test will be allowed in lieu of running the test. Such documentation shall be subjected to the approval of the procuring activity.

4.2.3 Qualification verification of production systems. In line 9 delete: "(WST33)" and replace with "(SY60)."

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4.3.1 Acceptance requirements. Delete and add: Acceptance test shall consist of those tests listed in table II of this specification sheet.

4.4.1.3.2 Control components. In first sentence, delete: "±5°C of the values specified by the detail specification" and replace with "±10°C of the values shown in figure 1 of MIL-E-81910."

Add: 4.4.3.10 Rated voltage. Voltage setting for the generator and regulator shall be as follows:

a. Generator: The rated voltage of the generator shall be 120V RMS, measured at the terminals of the generator.

b. GCU: The GCU shall be set to operate within the steady-state voltage regulation band as specified by figure 4 of this specification sheet.

4.5.3.1 Qualification test. Change "figure 3" to "figure 4 of this specification sheet."

4.5.3.2 Acceptance tests. Change in two places "figure 3" to "figure 4 of this specification sheet."

4.5.9 Efficiency. Add: During the test, the generator shall furnish control power, as specified in paragraph 3.5.5 of this specification sheet. Minimum efficiency with full, unity power factor load shall be 84.5 percent.

4.5.11 Output-voltage modulation. Delete and add: The output voltage modulation shall be determined when the generator is operated at minimum, average, and maximum rated speeds both at no load and rated current at rated voltage. The modulation shall be within the limits of figure 6.

4.5.12 Electromagnetic interference. Delete and add: See paragraph 3.5.8 of this specification sheet.

4.5.13 Waveform. Delete and add: The crest factor and harmonic content line-to-line and line-to-neutral of the output voltage shall be determined under the following conditions:

a. At minimum rated speed specified on the applicable drawing for no load, 50 percent rated current, and rated current unity power factor.

b. Condition a. shall be repeated for minimum rated power factor.

c. Conditions a. and b. shall be repeated at maximum rated speed.

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The line-to-neutral crest factor obtained by the above test shall be 1.414 ± 10 percent, and the observed crest factor from no load to rated load at any speed shall not exceed 10 percent. The value of any harmonic observed during the above test shall not exceed 3 percent of the fundamental.

4.5.14 Endurance. Delete and add: The generator system shall operate for a minimum time of 500 hours. The system shall be operated for the minimum time specified below for each test condition. Blast cooling air temperature in accordance with figure 2 of MIL-E-81910.

	Operating Time (Hrs) at Temperature (±5.5°C)	% Load	Altitude (Feet)	Total Time (Hrs)
a.	20 hrs at 93°C; 20 hrs at 25°C; 10 hrs at -55°C	0	sea level	50
b.	60 hrs at 93°C; 60 hrs at 25°C; 30 hrs at -55°C	50	sea level	150
c.	72 hrs at 93°C; 72 hrs at 25°C; 36 hrs at -55°C	100	sea level	180
d.	16 hrs at 40°C; 16 hrs at 0°C; 16 hrs at -40°C	50	50,000	48
e.	32 hrs at 40°C; 32 hrs at 0°C; 16 hrs at -40°C	100	50,000	80

Add: 4.5.14.1 Overload endurance. The generator system shall also withstand 100 applications of overloads of 200 percent at 0.75 power factor for 5 seconds, and 100 applications of overloads of 150 percent at 0.75 power factor for 2 minutes. These overloads shall be evenly distributed throughout the operating test time specified in 4.5.14.

Add: 4.5.14.2 Short circuits. The generator system shall sustain, without damage, the three short circuit fault conditions of essential zero impedance, line-to-line, and line-to-neutral faults. These faults shall be induced at the output of the generator once during each of the 12 load conditions specified in 4.5.14b thru e. for a total of 36 faults.

4.5.19 Generator fault torque. Delete second paragraph and replace with the following: "When the fault impedance producing maximum torque has been determined, it is to be applied for a minimum of 1.8 seconds with transient and steady-state torques recorded. No protective functions shall operate during the 1.8 second fault test."

4.5.20.1 Delete and add: Salt fog, control components. Control components shall be subjected to a salt fog test in accordance with MIL-E-81910. At the completion of this test each unit shall successfully demonstrate its buildup characteristics, the effects of application and removal of rated and 200 percent load, and protective functions.

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4.5.20.2 Delete and add: Oil-salt water ingestion, generators. The generator shall be subjected to an oil-salt water ingestion test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics and the effect of application and removal of rated and 200 percent load.

4.5.21 Fungus resistance. Delete and add: System components shall be subjected to a fungus test in accordance with MIL-E-81910, with the installation instructions and mounting nuts attached. At the completion of this test, each unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200 percent load, and protective functions. Mounting nuts are to be unaffected and the package holding the nuts suitable for shipment. Instructions are to be legible.

4.5.22 Sand and dust. Delete and add: The generator shall be subjected to a sand and dust test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics and the effect of application and removal of rated and 200 percent load.

4.5.23.1.1 Generator. Delete and add: The generator shall be subjected to a vibration test in accordance with MIL-E-81910, except that the vibration test curve contained in figure 5 of this detail specification will replace figure 3 of MIL-E-81910. The generator shall be mounted using the QAD adapter with the constant speed drive/starter unit attached.

Add: 4.5.23.1.1.1 Amplitude of vibration limit. The amplitude of vibration monitored at the anti-drive end of the generator during the main bending mode shall be limited to 20 g's but the vibration input shall not be lowered below 5 g's.

4.5.23.1.2 Control components. Delete and add: Control components shall be subjected to a vibration test in accordance with MIL-E-81910 except no random vibration required. At the completion of this test each unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200 percent load and protective functions.

4.5.24 Shock. Delete and add: Systems components shall be subjected to a shock test in accordance with MIL-E-81910. At the completion of procedure I, each unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200 percent load, the protective functions. There shall be no failure of the mounting attachments and the test item shall remain in place and not create a hazard during procedure III.

4.5.25 Humidity. Delete and add: System components except those tested under paragraph 4.5.20.2, shall be subjected to a humidity test in accordance with MIL-E-81910 except equipment will not be operated during humidity exposure. At the completion of this test each unit shall

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successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200 percent load, and protective functions.

4.5.30 Acceleration. Delete and add: The generator, in a non-operating condition, shall be subjected to the static loads of the linear accelerations specified in 3.4.2.8.1.2. After application of these loads, the generator shall be operated to determine that it suffered no adverse effects.

4.5.31 Disassembly and inspection. Delete everything after the first sentence.

Add: 4.5.34 Temperature and altitude. Test the system to requirements of paragraph 3.4.2.1 of this specification sheet.

Add: 4.5.35 Explosion and fire hazard test. Test in accordance with MIL-STD-810 Method 511.1.

6.3 Ordering data. Add: The acquiring activity will specify the applicable levels of packaging and packing.

6.4 Qualification. In line 8, delete "(Code WST33)" and replace with "(Code SY60)."

NOTES

1. Dimensions are in inches. Unless otherwise specified, tolerances: linear $\pm$0.030, angular $\pm 1^{\circ}$. All outline dimensions are maximum dimensions.

2. For design feature purposes, this standard takes precedence over acquisition documents referenced herein. Reference documents shall be of the issue in effect on date of invitation for bid.

Preparing Activity:
Navy - AS
(Project No. 6115-N477)

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TABLE I. Qualification tests.

Inspection	Para.	Generator	GCU	Cooling Condition
Examination of product	4.5.1	X	X	
Maximum speed for regulation	4.5.2		X	B
System control	4.5.3.1	X	X	A
Overspeed	4.5.4.1	X		A
Dielectric strength	4.5.5	X	X	
Phase balance	4.5.8.1	X	X	B
Efficiency	4.5.9	X	X	A
Output-voltage modulation	4.5.11	X	X	B
Electromagnetic interference	4.5.12	X	X	B
Waveform	4.5.13	X	X	B
Endurance	4.5.14	X	X	A
Operating position	4.5.15	X		B
Switching position	4.5.16	X	X	
Protective operation	4.5.17	X	X	A
Standard performance	4.5.18		X	A
Generator fault torque	4.5.19	X	X	A
Salt fog	4.5.20.1		X	
Oil-salt water ingestion	4.5.20.2	X		
Fungus resistance	4.5.21	X	X	
Sand and dust	4.5.22		X	

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TABLE I. Qualification test (continued).

Inspection	Para.	Generator	GCU	Cooling Condition
Vibration	4.5.23.1.1	X		
Vibration	4.5.23.1.2		X	B
Shock	4.5.24	X	X	
Humidity	4.5.25	X	X	
Altitude rating chart	4.5.29	X	X	
Acceleration	4.5.30	X		
Disassembly and inspection	4.5.31	X	X	
Temperature and altitude	4.5.34	X	X	
Explosion and fire hazard	4.5.35	X		

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TABLE II. Acceptance tests.

Inspection	Para.	Generator	GCU
Examination of product	4.5.1	X	X
Maximum speed for regulation	4.5.2	X	X
System control	4.5.3.2	X	X
Overspeed	4.5.4.2	X	
Dielectric strength	4.5.5	X	X
Phase balance	4.5.8.2	X	
Short-circuit capacity	4.5.10	X	X
Output-voltage modulation	4.5.11	X	X
Standard performance	4.5.18		X

NOTE: Acceptance tests shall be performed according to approved procedures and documentary evidence of compliance shall be submitted prior to qualification testing.

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